

Keduse Worku

Data Scientist | Applied Machine Learning | Statistical Modeling

773-595-2241 | keduseworku@aya.yale.edu | [keduseworku.github.io](https://github.com/keduseworku) | [linkedin.com/in/keduseworku](https://www.linkedin.com/in/keduseworku)

SUMMARY

PhD-trained data scientist specializing in statistical modeling and applied machine learning for complex systems. Experienced building probabilistic models, simulation pipelines, and data workflows in Python and SQL. Currently working on AI model evaluation at Anthropic while finishing a PhD in astrophysics at Johns Hopkins (2026).

EDUCATION

Johns Hopkins University

Ph.D. in Astrophysics

Baltimore, MD

Summer 2026

Yale University

Bachelor of Science (B.S.) in Astrophysics

New Haven, CT

2022

TECHNICAL SKILLS

Python, SQL, Machine Learning, Bayesian Modeling, Simulation, PyTorch, XGBoost, pandas, NumPy, JAX

RELEVANT EXPERIENCE

AI Model Evaluation Contractor

February 2026 - Present

Anthropic

- **Develop physics-based evaluation benchmarks** for testing multi-step reasoning in frontier large language models.
- **Analyze Claude Opus model** to identify reasoning breakdowns and **failure modes** in complex quantitative tasks.

Quantitative Research Summer Intern - Assets & Liabilities

June 2025 - August 2025

New York Life Insurance Company

New York, NY

- **Implemented novel callable bond pricing models** under the Hull-White short-rate framework via stochastic differential equations and trinomial trees, **reducing runtimes for fixed-income portfolio evaluations**.
- Developed and validated numerical interpolation strategies and unit tests for fixed-income pricing workflows.

Graduate Research Assistant

August 2022 - Present

Johns Hopkins University

Baltimore, MD

- Developed GPU-accelerated JAX simulations for neutrino-dark matter models, reducing runtime **from 1 hour to 10 seconds (600× speedup)**.
- **Built reproducible analytics pipelines** in Python/SQL for **large datasets (10,000+ records)**, improving data quality checks and downstream model reliability.
- Developed statistical models and validation workflows to ensure stable performance and reliable inference, **improving selection accuracy by 20% through uncertainty-aware modeling**.

LSSTC Data Science Fellow

September 2023 - August 2025

Multi-Institutional

Baltimore, MD

- Applied **ML methods (tree-based models, PCA)** for **anomaly detection** on large observational datasets and **built scalable pipelines for time-series processing**.

Predoctoral Researcher

January 2022 - August 2022

Princeton University

Princeton, NJ

- Built an end-to-end Python pipeline for **large-scale matrix computations and cross-correlation analysis**, improving parameter estimation efficiency and uncertainty quantification within a **100+ member collaboration**.
- **Engineered novel** contamination reduction **techniques**, enhancing resolution **by over 50%** for observational datasets.

Computational Research Intern

June 2021 - August 2021

Flatiron Center for Computational Astrophysics

New York, NY

- Optimized cross-correlation pipelines using **frequency-specific window functions**, reducing systematics and improving precision by **20%**.
- Derived multi-frequency statistical constraints that improved parameter estimation accuracy.

SELECTED APPLIED PROJECTS

- **Inventory Forecasting & Decision Making:** Built predictive forecasting models (XGBoost + Bayesian), translating demand uncertainty into order decisions and reducing expected cost versus heuristic approaches.
- **Latent State Inference for Financial Time Series:** Compared Bayesian HMC and Hidden Markov Models to infer latent market regimes from noisy price and volatility data, identifying early indicators of market stress.